8. Future of the Future

1 Unfortunately, the only kind of open universe model that will allow this is one without a cosmological constant, so even this tiny spark of hope seems to have been snuffed out by the current data.

2 Amazingly, Dyson himself never submitted his own paper. It was submitted to *Reviews of Modern Physics* on his behalf by a friend, who didn't ask permission. Dyson told me recently, "I didn't think it was worth publishing," considering it not to be appropriate for the journal. "It's always a matter of opinion," he added.

3 "See that star there? That star is going to EXPLODE in one year. Plus or minus four months. Just watch, you'll see." (Adapted liberally from Treu et al. 2016, *The Astrophysical Journal*.)

4 Thanks, Madrid-based theorist and CERN Scientific Associate José Ramón Espinosa. Very helpful.

5 Probably closer to 10^{15}, but I have a moral objection to the word "quadrillion."

6 This method is especially undesirable, Espinosa points out, since it "will not teach us anything, because we won't even see it coming."

7 Another thing Sean Carroll pointed out to me is that if his interpretation of quantum mechanics is correct, there are countless copies of ourselves in parallel universes that are at this very moment succumbing to vacuum decay. So he was probably never really going to be the best place to turn for relief from existential crisis.